

Factoring – “Bottoms Up” Method

If a Trinomial of the form $ax^2 + bx + c = 0$ is factorable, it can be completed using the Bottoms Up Method according to the following steps...

- Step 1.** Make sure the trinomial is in standard form ($ax^2 + bx + c = 0$).
- Step 2.** Factor out a GCF (greatest common factor) if applicable.
- Step 3.** Multiply $a \cdot c$ and re-write the polynomial as: $1x^2 + bx + ac = 0$.
- Step 4.** Factor as normal, by finding the two factors (n_1, n_2) of $a \cdot c$ that add up to b .
- Step 5.** Write the binomial factors as $(x + n_1)(x + n_2) = 0$.
- Step 6.** Divide the constants (n_1 and n_2) in each binomial factor by the original value of a .
- Step 7.** Simplify the resulting 2 fractions if applicable.
- Step 8.** If the simplified fraction has a denominator other than 1, move the denominator to become the coefficient in front of the variable (“bottoms up”).
- Step 9.** Check the answer - Multiply the answers to verify that you get the original trinomial.

Example 1

- Step 1: $6x^2 + 5x - 4 = 0$
- Step 2: No GCF
- Step 3: $a \cdot c = (6)(-4) = -24$

Re-write $\rightarrow x^2 + 5x - 24 = 0$

- Step 4: Find factors of -24
That add to b (5)

Factors $\rightarrow (+8)(-3)$

- Step 5: $(x + 8)(x - 3) = 0$

- Step 6: $\left(x + \frac{8}{6}\right)\left(x - \frac{3}{6}\right) = 0$

Divide the constants by the original value of a

- Step 7: $\left(x + \frac{4}{3}\right)\left(x - \frac{1}{2}\right) = 0$

Reduce the resulting fractions

- Step 8: $(3x + 4)(2x - 1) = 0$

Move the denominator so that it becomes the coefficient in front of the variable – “bottoms up”

Example 2

- $6x^2 - 21x - 45 = 0$
- $3(2x^2 - 7x - 15) = 0$
- $a \cdot c = (2)(-15) = -30$

Re-write $\rightarrow x^2 - 21x - 30 = 0$

- Find factors of -30
That add to b (-7)

Factors $\rightarrow (-10)(3)$

- $3(x - 10)(x + 3) = 0$

- $3\left(x - \frac{10}{2}\right)\left(x + \frac{3}{2}\right) = 0$

- $3(x - 5)\left(x + \frac{3}{2}\right) = 0$

- $3(x - 5)(2x + 3) = 0$

Step 9: $6x^2 + 5x - 4 = 0$ ✓

$6x^2 - 21x - 45 = 0$ ✓

“Bottoms Up” Factoring - Practice Problems

Directions - Factor the following trinomials by using the “bottoms up” factoring method.

<u>Problem</u>	<u>Answer</u>
1. $2x^2 - 9x - 18 = 0$	
2. $8x^2 + 2x - 3 = 0$	
3. $3x^2 + 19x = 40$	
4. $8x^2 - 12x - 8 = 0$	
5. $10x^2 - 25x = 125$	
6. $\frac{5}{2}x^2 - \frac{11}{2}x + 1 = 0$	